

Department of Computer and Software Engineering- ITU
CE100L: Computing Fundamentals and Programming Lab

Course Instructor: Nadir Abbas	Dated:
Lab Engineer: Muhammad Kashif	Semester: Fall 2025
Session: 2025-2029	Batch: BSCE2025

Lab 1. Getting Familiar with GitHub, Cygwin & Word Problems

Name	Roll number	Obtained Marks/35

Checked on: _____

Signature: _____

Objective

The goal of this handout is to learn what a GitHub, Cygwin is and explore problem-solving techniques.

Equipment and Component

Component Description	Value	Quantity
Computer	Available in lab	1

Conduct of Lab

1. Students are required to perform this experiment individually.
2. In case the lab experiment is not understood, the students are advised to seek help from the course instructor, lab engineers, assigned teaching assistants (TA) and lab attendants.

Theory and Background

GitHub is a web-based platform used for version control. Git simplifies the process of working with other people and makes it easy to collaborate on projects. GitHub Desktop is an application that enables you to interact with GitHub using a GUI instead of the command line or a web browser. You can push to, pull from, and clone remote repositories with GitHub Desktop.

Cygwin is a Unix-like environment and command-line interface for Microsoft Windows. Cygwin's purpose is expressed in its motto: "Get that Linux feeling – on Windows".

A **word problem** in math's is a math's question written as one sentence or more that requires children to apply their math's knowledge to a 'real-life' scenario.

The **C++ program** is written using a specific template structure. Which includes documentation, link section, definition section, global declaration section, function declaration section, main function.

Tasks

Task 1: Sign up an account with GitHub, keep your roll number as username e.g., BSCE240XX.

(<https://github.com/signup>).

Task 2: Accept the assignment posted in Google Classroom and after accepting clone the repository to your computer for this ensure you have logged into **github application**(installed in lab computers) with your account.

Task 3: Go to the start menu, search and open the Cygwin Terminal. Run the following commands to get the output of the code:

```
>>cd AddressOfYourClonedFolder (you can paste it by dragging your cloned folder in Cygwin)
>>make run
```

Task 5: Solve the given problems, answer using correct option (A, B, C, D, or E).

Task 6: Commit and Push the changes through the Github App

Assessment Rubric for Lab

Performance metric	CLO	Able to complete the task over 80% (4-5)	Able to complete the task 50-80% (2-3)	Able to complete the task below 50% (0-1)	Marks
1. Realization of experiment	1	Executes without errors excellent user prompts, good use of symbols, spacing in output. Through testing has been completed.	Executes without errors, user prompts are understandable, minimum use of symbols or spacing in output. Some testing has been completed.	Does not execute due to syntax errors, runtime errors, user prompts are misleading or non-existent. No testing has been completed.	
2. Conducting experiment	1	Able to make changes and answered all questions.	Partially able to make changes and few incorrect answers.	Unable to make changes and answer all questions.	
3. Computer use	2	Document submission timely.	Document submission late.	Document submission not done.	
4. Teamwork	3	Actively engages and cooperates with other group member(s) in effective manner.	Cooperates with other group member(s) in a reasonable manner but conduct can be improved.	Distracts or discourages other group members from conducting the experiment	
5. Laboratory safety and disciplinary rules	3	Code comments are added and does help the reader to understand the code.	Code comments are added and does not help the reader to understand the code.	Code comments are not added.	
6. Data collection	3	Excellent use of white space, creatively organized work, excellent use of variables and constants, correct identifiers for constants, No line-wrap.	Includes name, and assignment, white space makes the program fairly easy to read. Title, organized work, good use of variables.	Poor use of white space (indentation, blank lines) making code hard to read, disorganized and messy.	
7. Data analysis	4	Solution is efficient, easy to understand, and maintain.	A logical solution that is easy to follow but it is not the most efficient.	A difficult and inefficient solution.	
Total (out of 35):					